

Lecture VII - NumPy for Scientific Computing

Programming with Python

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Episode 7: The Numbers Deck

The numbers deck is due

The investor wants a numbers deck. **Two weeks of orders** sit in the system, one row each: minutes, price, zone.

...

Tobi's plan is a **40-tab spreadsheet**, one tab per zone, copied by hand. He looks at the pile, then at his list-of-lists, and says the only true thing he'll say all week:

"We need a bigger boat than a list."

...

Today we get the bigger boat: **NumPy**, one object that holds a thousand numbers and does arithmetic on all of them at once.

Warm-up

Question 1

Three questions from Episode 6. Commit. Hands up **before** the reveal.

```
from math import ceil

# which line works?
```

a) `ceil(3.2)` b) `math.ceil(3.2)` c) both

Answer 1

a) **`ceil(3.2)`**: `from math import ceil` binds only the **name** `ceil`. The module `math` itself was never imported, so `math.ceil` has nothing to reach through.

Question 2

```
import random
```

```
random.seed(42)
print(random.randint(1, 20))
print(random.randint(1, 20))
print(random.randint(1, 20))
```

Tobi runs this exact script today and again tomorrow. Tomorrow's numbers are...

a) different: random is random b) an error: seed 42 was already used c) the same three numbers

Answer 2

c) the same three numbers. Every run starts from seed 42, so the stream replays from the top. That is the entire job of a seed: reproducible randomness. (Two batches *inside one run* would differ: the stream continues; a fresh run rewinds it.)

Question 3

```
import statistics

statistics.median([9, 2, 5])
```

returns...

a) 2 b) 5 c) an error: the list is not sorted

Answer 3

b) 5: median sorts the values internally before picking the middle one. You never have to sort first; 5 is the middle of 2, 5, 9.

A thousand orders at once

From a list to an array

A **NumPy array** holds many numbers under one name, like a list, but built for math. You make one from a list, then ask it about itself:

```
import numpy as np

prices = np.array([12.0, 9.0, 15.0])
print(prices)
print(prices.shape)    # how many, in each dimension
print(prices.dtype)    # what kind of number
print(prices.size)     # how many in total
```

```
[12.  9. 15.]
(3,)
float64
3
```

...

`np.array(...)` wraps a list; `import numpy as np` is the nickname everyone uses. `.shape` is `(3,)`, `.dtype` is `float64`, `.size` is `3`. One `dtype` for the **whole** array: every element shares the same type; that's part of what makes it fast.

One operation, every element

Here's the bigger boat. The deck needs **gross** prices: 19% VAT on all three. With a list you loop; with an array you just multiply:

```
# the painful way: a loop, item by item
gross = []
for p in [12.0, 9.0, 15.0]:
    gross.append(p * 1.19)
```

```
import numpy as np

prices = np.array([12.0, 9.0, 15.0])
print(prices * 1.19)  # every element, one expression
```

```
[14.28 10.71 17.85]
```

...

One operation lands on **all** elements at once: `[14.28 10.71 17.85]`. No loop, no `.append`, and on a thousand orders it's also far faster.

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_a/



First **predict** what happens, then run it.

Arrays from scratch

Two builders make evenly-spaced arrays without typing every number, handy for axes and ranges:

```
import numpy as np

print(np.arange(0, 10, 2))    # start, stop (excluded), step
print(np.linspace(0, 1, 5))  # start, stop (included), how many
```

```
[0 2 4 6 8]
[0.  0.25 0.5  0.75 1.  ]
```

...

`arange` walks by a **step** and stops before the end, just like `range`. `linspace` splits a span into a fixed **count** of points, endpoints included.

Predict: times two

Tobi multiplies a row of counts by two. What does this print?

```
print([1, 2, 3] * 2)
```

a) `[1, 2, 3, 1, 2, 3]` b) `[2, 4, 6]` c) an error

...

Predict first. Pick a letter, then I reveal the answer.

Answer: lists repeat, arrays compute

a) `[1, 2, 3, 1, 2, 3]`. That's a plain **list**, and `* 2` on a list *repeats* it. Wrap it in an array and the same `* 2` does the math instead:

```
import numpy as np

print([1, 2, 3] * 2)          # list → repeated
print(np.array([1, 2, 3]) * 2) # array → doubled
```

```
[1, 2, 3, 1, 2, 3]
[2 4 6]
```

...

Lists repeat; arrays compute. That's why we're here.

Predict: plus

Tobi adds this week's zone counts to last week's. What does this print?

```
print([1, 2, 3] + [10, 20, 30])
```

a) `[11, 22, 33]`, added pairwise b) an error: lists don't add c) `[1, 2, 3, 10, 20, 30]`

...

Predict first. Pick a letter, then I reveal the answer.

Answer: lists glue, arrays add

c) `[1, 2, 3, 10, 20, 30]`. On lists, `+` **glues** them end to end, exactly like `* 2` repeated. On arrays, `+` adds element by element, first with first, second with second:

```
import numpy as np

print([1, 2, 3] + [10, 20, 30])          # list → glued
print(np.array([1, 2, 3]) + np.array([10, 20, 30])) # array → added
```

```
[1, 2, 3, 10, 20, 30]
[11 22 33]
```

...

Every operator follows the same rule: lists do list things, arrays do math.

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:



First **predict** what happens, then run it.

Asking questions of data

A comparison makes a mask

Compare an array to a number and you don't get one `True/False`. You get a **whole array** of them, one per element. That's a **mask**, and you can filter with it:

```
import numpy as np

times = np.array([25, 41, 18, 33])
print(times > 30)           # a True/False for every element
print(times[times > 30])    # keep only where the mask is True
```

```
[False  True False  True]
[41 33]
```

...

`times > 30` is the mask; `times[times > 30]` reads the array **through** the mask and returns just the matching values. No loop, no `if`.

Counting without a loop

A mask answers two investor questions at once. `.sum()` counts the `Trues` (each counts as 1); `.mean()` gives the **share** that are `True`:

```
import numpy as np

times = np.array([25, 41, 18, 33, 52, 29, 44, 12])
```

```
late = times > 30
print(int(late.sum()))           # how many were late
print(float(times[late].mean())) # average of just the late ones
print(float(late.mean()))       # the SHARE that were late
```

```
4
42.5
0.5
```

...

4 deliveries over 30 minutes, averaging 42.5, and `late.mean()` says **half** the run was late, one line each, straight into the deck.

Predict: counting the Trues

What does calling `.sum()` on the mask give?

```
print((np.array([1, 5, 3]) > 2).sum())
```

a) True b) 2 c) [False, True, True]

...

Predict first. Pick a letter, then I reveal the answer.

Answer: True counts as 1

b) 2: `.sum()` adds the mask up, and each **True** is worth 1, each **False** 0. Two elements clear the bar, so the count is 2:

```
import numpy as np

print(np.array([1, 5, 3]) > 2)           # [False True True]
print((np.array([1, 5, 3]) > 2).sum())   # Trues add up to 2
```

```
[False True True]
2
```

...

Summing a mask counts; averaging a mask shares. Same two tricks the lab asks for.

Predict: two conditions

The investor wants deliveries that took **more than 20 and less than 40** minutes. Tobin writes what he'd write for two numbers:

```
times = np.array([25, 41, 18, 33])
print(times > 20 and times < 40)
```

a) an error: ambiguous truth value b) `[ True False False True]` c) `True`, both sides hold somewhere

...

Predict first. Pick a letter, then I reveal the answer.

Answer: `and` wants one truth, a mask has four

a) **an error.** `and` asks the left side “are you true?”, and a four-element mask has no single answer. Between two masks use `&`, and wrap each side in parentheses (`&` binds tighter than `>`):

```
import numpy as np

times = np.array([25, 41, 18, 33])
print(times > 20 and times < 40)
```

```
ValueError: The truth value of an array with more than one element is
ambiguous. Use a.any() or a.all()
```

```
[?] [31m-----[?] [39m
```

```
[?] [31mValueError[?] [39m                                Traceback (most recent
call last)
```

```
[?] [36mCell[?] [39m[?] [36m [?] [39m[?] [32mIn[9][?] [39m[?] [32m, line 4[?] [39m
[?] [32m                                1[?] [39m [?] [38;5;28;01mimport[?] [39;00m[?] [38;5;250m
[?] [39m[?] [34;01mnumpy[?] [39;00m[?] [38;5;250m
```

```
[?] [39m[?] [38;5;28;01mas[?] [39;00m[?] [38;5;250m [?] [39m[?] [34;01mnp[?] [39;00m
[?] [32m                                3[?] [39m times = np.array([?] [32m25[?] [39m, [?] [32m41[?] [39m,
[?] [32m18[?] [39m, [?] [32m33[?] [39m])
```

```
[?] [32m----
> [?] [39m[?] [32m4[?] [39m [?] [38;5;28mprint[?] [39m([?] [43mtimes[?] [49m[?] [43m
[?] [49m[?] [43m>[?] [49m[?] [43m                                [?] [49m[?] [32;43m20[?] [39;49m[?] [43m
[?] [49m[?] [38;5;129;43;01mand[?] [39;49;00m[?] [43m [?] [49m[?] [43mtimes[?] [49m[?] [43m
[?] [49m[?] [43m<[?] [49m[?] [43m [?] [49m[?] [32;43m40[?] [39;49m)
```

```
[?] [31mValueError[?] [39m: The truth value of an array with more than one element
is ambiguous. Use a.any() or a.all()
```

```
print((times > 20) & (times < 40))    # element by element: & combines masks
```

```
[ True False False  True]
```

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_c/



First **predict** what happens, then run it.

The days-by-zones grid

A grid of numbers

Real data isn't one row. Stack rows and you get a **2D array**: here three days (rows) across four zones (columns: Nord, Sued, Hafen, Altstadt):

```
import numpy as np

deliveries = np.array([[ 9, 14, 11,  6],
                      [15, 12,  8,  9],
                      [13, 20, 16, 11]])

print(deliveries.shape)      # (rows, columns) → (3, 4)
print(deliveries[0, 2])      # row 0, column 2
```

```
(3, 4)
11
```

...

`.shape` is now `(3, 4)`: three days, four zones. One index picks the **row**, a second the **column**: `deliveries[0, 2]` is day 0, zone 2 (Hafen).

Predict: the shape

Tobi builds a grid with two more days. What does `.shape` say?

```
grid = np.array([[ 9, 14, 11,  6],
                  [15, 12,  8,  9],
                  [13, 20, 16, 11],
                  [10, 17, 12,  8],
                  [ 7, 11,  9,  5]])
print(grid.shape)
```

a) (20,) b) (5, 4) c) (4, 5)

...

Predict first. Pick a letter, then I reveal the answer.

Answer: rows first, then columns

b) (5, 4). `.shape` is always (rows, columns): five inner lists make five rows, each with four numbers. (20,) would be one flat row of twenty; 20 is `.size`, the total count:

```
import numpy as np

grid = np.array([[ 9, 14, 11,  6],
                  [15, 12,  8,  9],
                  [13, 20, 16, 11],
                  [10, 17, 12,  8],
                  [ 7, 11,  9,  5]])

print(grid.shape)    # (rows, columns)
print(grid.size)     # rows × columns
```

```
(5, 4)
20
```

...

Same order as indexing: `grid[row, column]`, rows first.

Which way to collapse?

To sum a grid you must say **which way** to collapse it. The `axis` tells NumPy which direction disappears:

	Nord	Sued	Hafen	Altstadt
day0	[9	14	11	6]
day1	[15	12	8	9]
day2	[13	20	16	11]

axis=0 collapses DOWN the rows:

↓ ↓ ↓ ↓

37 46 35 26 one number per column (zone)

axis=1 collapses ACROSS the columns:

day0 → 40 · day1 → 44 · day2 → 60 one number per row (day)

axis in code

The same grid, the same two collapses, one keyword each:

```
import numpy as np

deliveries = np.array([[ 9, 14, 11,  6],
                       [15, 12,  8,  9],
                       [13, 20, 16, 11]])
print(deliveries.sum(axis=0))  # DOWN the rows → per zone
print(deliveries.sum(axis=1))  # ACROSS the columns → per day
```

```
[37 46 35 26]
[40 44 60]
```

...

axis=0 collapses DOWN the rows, one number per column (zone): [37 46 35 26].

axis=1 collapses across, one per day: [40 44 60].

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_d/



First **predict** what happens, then run it.

Where does the max sit?

The per-zone totals answer “how many?”, but the investor asks “**which** zone?”. `argmax` tells you **WHERE** the maximum sits, as an index:

```
import numpy as np

deliveries = np.array([[ 9, 14, 11,  6],
                      [15, 12,  8,  9],
                      [13, 20, 16, 11]])
zone_totals = deliveries.sum(axis=0) # [37 46 35 26]
print(int(zone_totals.argmax()))    # index of the biggest
```

```
1
```

...

`max` would give the value 46; `argmax` gives its **position**, 1. The busiest zone is **Sued**, sitting at index 1, not at the front. Position, not value: that’s the whole point of `argmax`.

Predict: which call collapses which way?

The investor wants **four** numbers, one per zone. Which call?

```
week = np.array([[ 9, 14, 11,  6],
                 [15, 12,  8,  9],
                 [13, 20, 16, 11]])
```

a) `week.sum(axis=1)` b) `week.sum()` c) `week.sum(axis=0)`

...

Predict first. Pick a letter, then I reveal the answer.

Answer: collapse the days, keep the zones

c) `week.sum(axis=0)`. Four zones means four numbers, so the **days** must disappear: `axis=0` collapses DOWN the rows, one number per column (zone):

```
import numpy as np

week = np.array([[ 9, 14, 11,  6],
                 [15, 12,  8,  9],
                 [13, 20, 16, 11]])
print(week.sum(axis=0))
```

```
[37 46 35 26]
```

...

a) would give **three** numbers (one per day); b) would give **one** number: the grand total, 144.

Your turn: 5-10 minutes

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_e/



First **predict** what happens, then run it.

To the Lab

After the break: the lab

- Head to the lab notebook: [Episode 7: The Numbers Deck](#)
- You'll turn the order log into arrays, add VAT to a whole price column at once, mask out the late deliveries to count and average them, and collapse a days-by-zones grid to find the winning zone
- AI is allowed, so try the chatbot, and keep your one-line disclosure note on the submission
- It runs entirely in your browser: no setup, just click and code

...

! Important

Download your `.py` before you leave. Closing the tab without downloading loses your work.

Wrap-up

Three things to remember

1. **One array, one operation.** `np.array([...])` holds many numbers; arithmetic hits **every element at once**. No loop. Ask it `.shape`, `.dtype`, `.size` to know what you're holding.
2. **A comparison makes a mask.** `arr > 30` is a True/False array: `arr[mask]` filters, `.sum()` counts the Trues, `.mean()` gives their share.
3. **In 2D, pick an axis.** `axis=0` collapses DOWN the rows (one number per column), `axis=1` across the columns (one per row); `argmax` tells you where the maximum sits.

...

Note

Next episode starts with **Checkpoint 4**: 40 minutes, AI allowed, everything from Episodes 6-7. Then the investor opens a data room, and Tobi lets an AI write his pandas: the arrays get column names, and eighty orders become a table you can query.

Literature

Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

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Note

NumPy has excellent free docs: the [NumPy absolute beginner's guide](#) covers everything in this session and a little more.

...

For more, see the [literature list](#) of this course.